
Team Control Number

16302

Problem Chosen

B

2024

HiMCM/MidMCM

Summary Sheet

High-performance computing (HPC) facilities are essential for a wide range of scientific and technological advancements; most notably in the field of artificial intelligence (AI). However, its ever-increasing environmental impact is an ever-increasing concern due to its immense energy consumption and reliance on non-renewable energy sources. Hence, this model focuses on the energy consumption of these facilities, most notably its environmental impact, as well as feasible methods to make these facilities more sustainable.

Hence, via our model, we have figured that the transition of HPCs to a fully renewable energy grid is ideal but impractical due to limitations in resources and computational power. A more feasible approach, however, involves the adoption of a hybrid energy model, wherein 30% of the energy is derived from renewable sources, and the remaining 70% comes from non-renewable sources. Thus, a feasible yet sustainable balance is created.

However, land availability is one of the key challenges to implementing renewable energy on HPC sites. Solar panels, one of the most viable renewable energy sources for most facilities, require a significant amount of land, limiting the energy harnessed. Even with optimizations, only 30% of the energy consumed by an HPC may (realistically) come from renewable sources. Despite this, the transition to renewable energy significantly reduces the carbon emissions associated with the HPCs' operations, demonstrating that even partial integration of renewable energy can have a meaningful environmental impact. Further energy efficiency improvements, such as implementing efficient algorithms, maximizing energy utilization rates, and adopting liquid cooling systems can further reduce energy consumption and environmental impact.

In conclusion, while a fully renewable energy model for HPCs is not feasible in the present, the integration of a mere 30% renewable energy, and the implementation of energy-efficient technologies offer a highly sustainable solution. By implementing these strategies, HPC facilities may lower their environmental impact, reduce environmental harm, and continue the support of vital computational needs. Through these efforts, the environmental sustainability of HPCs shall be significantly enhanced, making them more aligned with global efforts to reduce carbon emissions altogether.

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1 | Introduction

The rise of high-powered computing (HPC) as a tool for faster and more efficient computations continues to grow as sectors such as artificial intelligence, data science, and cryptocurrency mining rapidly develop. HPC allows for quadrillions of complex computations on data within seconds, becoming a vital breakthrough in all its applied industries, such as running large AI models using unstructured data sets. The use of HPC facilities has been completely unavoidable following their implementation.

High-powered computing goes far beyond the limits of typical modern computers. Not only is HPC fast, but it is also scalable, being able to scale up or allocate a fraction of the facility to match the complexity and amount of data being handled for any task it is given. Overall, this speed and computational efficiency make the usage of an HPC facility relatively more cost-effective than previous methods of computing large amounts of data. That is why HPC facilities are implemented worldwide at such a large scale.

With the high level of computational power of HPC and the massive scale of its usage, its utilization consumes an enormous amount of energy. The high demand for HPC and insane energy requirements call for an energy quota that cannot (currently) be met sustainably. The abundant use of fossil fuels to power HPC facilities has led to an obvious correlation of HPC to rising environmental problems like greenhouse gases and carbon emissions. HPC facilities now contribute to about 2% of all carbon emissions in the world with annual energy consumption growing by 10-12% by the year.

The adverse effects of HPC facilities don't end there. HPC facilities produce tons of air pollution during their lifetime and maintenance. This negatively affects the health of people and ecosystems located near these facilities, their energy sources, and their disposal sites. For these reasons and more, HPC facilities must transition to the use of renewable energy sources, but that transition will be far from cheap.

HPC facilities are undeniably necessary, but big changes need to be made to justify their utilization by ensuring a sustainable, safe, and cost-effective way to apply HPC. Our team has developed a detailed model that aims to examine the effects of converting into a 100% renewable energy utilization for HPCs in terms of energy usage, carbon emissions, and other important factors.

2 | Model and Solution

High-performance computing (HPC) plays a pivotal role in technology, which is ingrained from the most advanced of discoveries to even our daily lives. However, the immense energy demands of HPC systems present significant environmental challenges, particularly concerning carbon emissions. Understanding and mitigating these environmental impacts requires a thorough understanding of the environmental impact that HPC facilities have on our environment. Hence, this solution introduces a comprehensive model to quantify and analyze the energy used by HPC energy consumption, as well as the carbon emissions that result from it.

In this solution, the terms HPC cluster and HPC facility will be used interchangeably. They will both refer to the logical unit for an HPC's computing power combined with the physical facility required to house the servers for computing. For the sake of simplicity, we will assume every HPC cluster has the same components, but not necessarily the same energy consumption for each component.

2.1 Variables and Definitions

Variable	Definition
I	Total environmental impact represented by carbon footprint in kilograms of carbon dioxide (kg CO ₂) for one year.
M	Weighted average emission factor in kilograms of carbon dioxide per kilowatt (kg CO ₂ /kWh) depending on the energy source.
P_m	Weighted average emission factor in kilograms of particulate matter per kilowatt (kg CO ₂ /kWh) depending on the energy source.
D_m	Weighted average emission factor in kilograms of dust per kilowatt (kg CO ₂ /kWh) depending on the energy source.
E_C	Energy consumption of one HPC cluster in kilowatt-hours (kWh) for one year.
E_F	Energy consumption of one Fossil Fuel Power Plant in kilowatt-hours (kWh) for one year
N_{HPC}	Number of HPC clusters.
N_{FFPP}	Number of Fossil Fuel Power Plants.
a	The average utilization rate of energy consumed by an HPC cluster in percentage (%).

Variable	Definition
src_m	The full capacity of energy output that a source of energy can produce under optimal conditions in kilowatt-hours (kWh) for one year.
src_a	The actual capacity of energy output that a source of energy can produce under actual conditions in kilowatt-hours (kWh) for one year.
E_w	Energy wasted
N_{src}	Number of energy sources.
N_{server}	Number of servers in an HPC cluster.
E_h	Energy consumption of all hardware in one HPC cluster in kilowatt-hours (kWh) for one year.
E_s	Energy consumption of all software in one HPC cluster in kilowatt-hours (kWh) for one year.
E_f	Energy consumption of a physical HPC cluster in kilowatt-hours (kWh) for one year (electricity, cooling).
e_s	Energy consumption of storage in one HPC cluster in kilowatt-hours (kWh) for one year.
e_n	Energy consumption of the network of one HPC cluster in kilowatt-hours (kWh) for one year.
e_{server}	Energy consumption of a server in one HPC cluster in kilowatt-hours (kWh) for one year.
e_e	Energy consumption of electricity in one HPC cluster in kilowatt-hours (kWh) for one year.
e_c	Energy consumption of a cooling system in one HPC cluster in kilowatt-hours (kWh) for one year.
N_{cooler}	Number of coolers required to cool one server.
e_{cooler}	Energy consumption of a cooler in kilowatt-hours (kWh) for one year.
s_{total}	Size of one HPC facility in square meters (m^2).
n_{solar}	Number of solar panels to power one HPC cluster.

Variable	Definition
n_{wind}	Number of wind turbines to power one HPC cluster.
C_{solar}	Power capacity of one solar panel in kilowatt-hours (kWh) for one year.
C_{wind}	Power capacity of one turbine in kilowatt-hours (kWh) for one year.
S_{solar}	The size of one solar panel in square meters (m^2).
S_{wind}	The size of one wind turbine in square meters (m^2).

2.2 Environmental Impact

A high-powered computing cluster is made up of many components, as seen in the figure below.

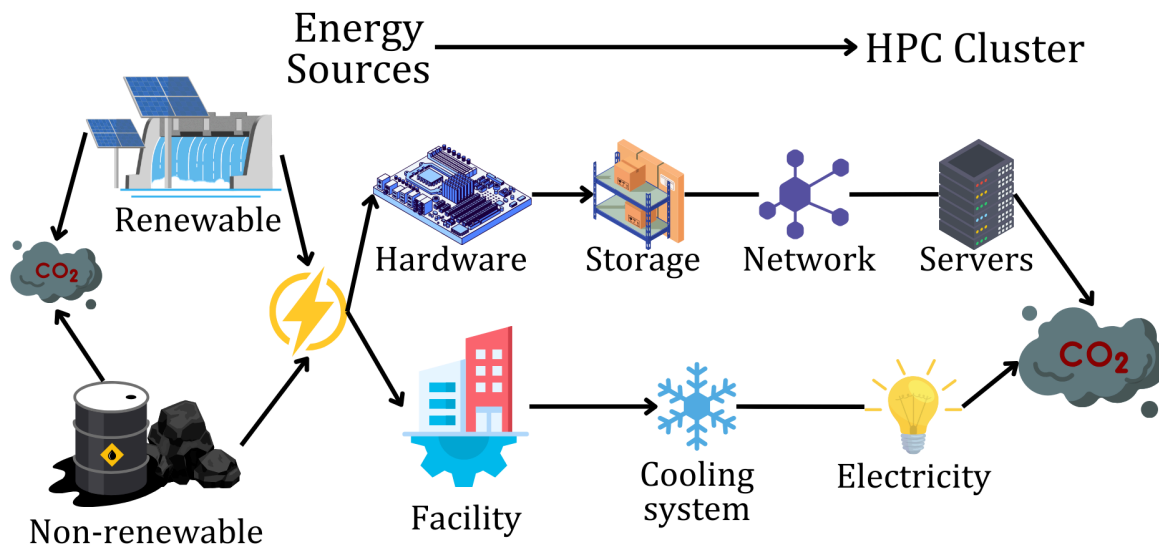


Figure 2.2.1. The aspects of a high-powered computing cluster that contribute to carbon emissions.

Our model will incorporate these specific components so that each required value can easily be measured physically on a small scale. It is important to break down large values so that they can be manually computed or measured in real life, which is why we decided to include all the details of an HPC cluster instead of just demanding a cumulative value for total energy consumed for one cluster. Additionally, this allows for more modifications in the model and a deeper understanding of what causes the large environmental footprint of HPC.

2.2.1 Environmental Impact of HPC from Worldwide Facilities

The total environmental impact of HPC from facilities or clusters for one year can be modeled by:

$$I = N_{HPC} E_C M$$

The energy consumption of one HPC cluster is given by:

$$E_C = E_h + E_f$$

where

$$E_h = e_s + e_n + N_{server} e_{server}$$

$$E_f = e_e + e_c$$

$$\text{and } e_c = N_{server} N_{cooler} e_{cooler}$$

The weighted average emission factor can be calculated based on the energy sources that power an HPC cluster. Determining this factor accounts for possible energy mixes in powering HPCs. It is given by:

$$M = \sum_{i=1}^{N_{src}} (P_i m_i)$$

where P_i is the proportion of energy from source i , and m is the emission factor of the same source. For example, if 30% of an HPC cluster is powered by solar energy and 70% is powered by natural gas, the weighted average emission factor for all the energy sources is

$$M = (0.3)(0.04) + (0.7)(0.16) = 0.124$$

since the emission factors of solar energy and natural gas are 0.4 and 0.16 kg CO₂/kWh, respectively.

Combining all the necessary information,

$$I = N_{HPC} (e_s + e_n + N_{server} e_{server} + e_e + N_{server} N_{cooler} e_{cooler}) \left(\sum_{i=1}^{N_{src}} (P_i m_i) \right)$$

2.2.2 Maximizing Energy Efficiency

Another consideration to keep in mind is that carbon emissions don't only come from the energy consumed by HPC clusters, but also from excess unutilized energy output by the source. The energy outputs can be modeled as follows:

$$src_m = \sum_{i=1}^{N_{src}} e_i,$$

where e_i is the energy output from source i in kilowatt-hours annually. This makes up the maximum output that a power plant or energy source can produce under optimal conditions. In this solution, we cannot assume that the energy sources are always running at full capacity, which we need to account for. For an energy source to be more effective with energy allocation and utilization, the average utilization rate of energy needs to be near 100%. This percentage is given by:

$$a = \frac{src_a}{src_m}$$

and shows how effectively a power plant or energy source is being used by a facility over a year. This average utilization rate is different for different types of energy sources such as coal and nuclear energy.

2.3 Application of the Model

2.3.1 The Growth of HPC

The number of individuals utilizing the internet is expected to correlate with the number of users leveraging cloud services, as it is known that most internet-based applications rely heavily on high-performance computing (HPC) solutions, such as cloud infrastructure. Furthermore, it is justifiable that there is a strong relationship between the growth of internet users and the HPC market size — such as the one in the United States — as evidenced by the similarity in their growth trends — a useful sanity check. In addition, from an economic perspective, as the demand for cloud services rises, so shall the supply, thus driving market expansion.

Finally, a larger HPC market size indicates an increase in the number of HPC facilities. Hence, this suggests a positive relationship between these two variables, though the relationship is likely not linear. However, there is room for discrepancy in our estimate as not every internet user directly contributes to one HPC cluster, hence a coefficient that dampens their effect (but not completely disregards it) would be highly helpful for the model.

Therefore, with these in mind, we decided to model the number of HPC facilities as an exponential function that takes into account the size of the market of HPCs, and the size of the population that makes use of the internet as our estimate of the population that makes use of

these HPC facilities. This is because the population growth of people who use the internet is exponential, and thus the number of HPC facilities is likely to follow a similar trend.

Therefore, we can approximate:

$$N_{HPC}(t) \approx ce^{c(t-2022)}$$

where $N_{HPC}(t)$ represents the number of HPC facilities as a function of time, t represents the year that we are interested in finding the number of HPC facilities for, and c as a coefficient, which is defined as the market size of HPCs in US dollars, multiplied by the number of internet users at year y divided by the scaling factor, set to 10^7 , which provides a relatively realistic number of HPC facilities. The coefficient c can be calculated using this [code](#), which is computed to be the following:

$$c = 10^{-7} \cdot 48.51 (1.075)^{t-2022} \cdot \frac{1.20 \times 10^{10}}{1 + e^{-0.1172(t-2024.35)}}$$

Although there is an ever-increasing demand for HPC facilities, given how there are finite resources on earth, the growth of the number of HPC facilities would eventually slow down thus our model would follow a somewhat logarithmic or logistical growth in the future. In addition, climate change initiatives may affect the growth of the number of HPC facilities as these facilities significantly contribute to pollution; however, given the short period between 2024 and 2030, this model shall serve as a fair estimate of the number of HPC facilities per year, which is computed to be roughly 68,320 in number for 2030. This is far worse than the number we computed for 2024, which is measly half of HPC facilities: 32,867. This implies that there is likely a significant increase in HPC facilities, and thus a substantial increase in the carbon footprint of said HPCs.

2.3.2 Looking into the Future

With this expected number of HPC facilities, by the least, the environmental impact of HPCs in 2030 would grow to twice the amount it has today, as it has a direct relationship with the expected number of HPC facilities by 2030; however, to calculate for the specific value:

$$I_{2030} = N_{HPC}(2030)E_c M$$

where E_c is defined to be

$$E_c = e_s + e_n + N_{server} e_{server} + e_e + N_{server} N_{cooler} e_{cooler}$$

and where M is defined to be

$$M = \sum_{i=1}^{N_{src}} (P_i m_i)$$

Finding the specific value I is incredibly difficult as there are various factors within each HPC facility, with generally little public data available for reference. Despite this, there is reasonable justification to conjecture that, as long as there are no significant technological advancements, the terms that proceed $N_{HPC}(2030)$ shall stay constant between 2024 and 2030. This is because these terms refer to internal properties of the HPC facilities rather than properties that rely on several HPC facilities. Hence, the only value that significantly changes with respect to time is $N_{HPC}(t)$, which has been modeled in the earlier part of this section.

It is sufficient for the computed value E_C to be derived from data from 2024 as it has been shown that E_C is not likely to change significantly within the short period as long as no significant changes to the mechanism of HPC facilities themselves are made. Hence, to compute E_C , the values of the corresponding terms are to be found from existing resources and calculations:

Variable	Estimated value
e_s	4400000 kWh
e_n	3700000 kWh
N_{server}	200
e_{server}	5500 kWh
e_e	13500 kWh
N_{cooler}	10
e_{cooler}	1000 kWh

From this,

$$\begin{aligned}
 E_C &= e_s + e_n + N_{server} e_{server} + e_e + N_{server} N_{cooler} e_{cooler} \\
 &= 4400000 + 3700000 + (200)(5500) + 13500 + (200)(10)(1000)
 \end{aligned}$$

$$= 11213500 \text{ kWh per year}$$

Confirming our model's estimate to existing records, a medium-sized HPC can consume from 0.5 MW to 5 MW of power annually. Assuming an HPC cluster is always active due to high demand, this is around 8760 hours per year. Converting from kWh to MW, 11213500 kWh = 1.28 MW. This fits within the range of expected values.

Then, to find M , it is sufficient to assume that in the status quo, most HPC clusters are using fossil fuels (particularly the most common source of electricity, coal) to power their facilities. Hence, M is derived from the following calculation:

$$M = (1.0)(0.27) = 0.27$$

Hence, combining the terms and thus attaining the value of I_{2030} :

$$I_{2030} = N_{HPC}(2030)E_C M$$

$$I_{2030} = (68320)(11213500)(0.27) = 206848706400 \approx 2.07 \times 10^{11} \text{ kg CO}_2$$

This is an alarmingly large number for the impact of HPC facilities by the year of 2030; in fact, if we try to calculate this for the year 2024:

$$I_{2024} = N_{HPC}(2024)E_C M$$

$$I_{2024} = (32867)(11213500)(0.27) = 99509608215 \approx 9.95 \times 10^{10} \text{ kg CO}_2$$

The value I_{2030} is more than double that of the value of I_{2024} , which is a risky environmental scenario considering that this is within six years. However, there is a caveat to this way of calculating I_{2030} , which is the fact that fossil fuels were the only source of energy considered. We assumed the worst case since fossil fuels are one of the major factors that cause the environmental impact of HPC facilities to balloon, and the factor that is also the most feasible to manage and modify in practice. Among fossil fuels, coal has the highest emission factor.

2.4 Transitioning to Renewable Energy

Ideally, the easiest way to reduce the problems faced by HPCs would be to transition to a fully renewable energy source of power. However, with resource constraints among other factors, transitioning into a 100% renewable energy model would be impossible.

One of the primary factors the group found to be leading the cause of the inability to transition is Land Availability. The amount of land available to set up these renewable energy sources is already limited by itself without considering the large number of HPCs already set up while being powered.

This section will model and prove the implications of the above-stated constraint of transitioning into a fully renewable energy source. Following this, the importance of renewable energy will be reestablished together with its effects on the environment and HPCs following certain limitations.

2.4.1 Land Availability

The amount of renewable energy stations that can be placed in one HPC facility of a variable power greatly depends on the amount of land available for utilization. Here, it is important to note that in the best case, the available amount of land for the entire facility is also the available amount of land for that given facility, and in the worst case, the only available land is the area surrounding the facility. Thus, the amount of land limits the amount of power generated by the energy source. Given that HPCs consume thousands of kilowatts each day (thus also leading to high carbon emissions), it is a fact that the amount of land to be utilized should be maximized to better reach a 100% renewable energy source.

This can be modeled as follows:

$$n_{solar}s_{solar} + n_{wind}s_{wind} \leq s_{total}$$

and

$$C_{solar}n_{solar} + C_{wind}n_{wind} \geq E_c$$

The scope of this model covers two types of renewable energy sources which are solar power and wind. The first inequality shows the maximization of the number of renewable energy sources given the area of a facility. This inequality essentially adds the areas of all renewable energy sources which ensures that it fits within the area of the total facility. However, an important limitation to note is that s_{total} may still vary as the years go on with more expansions of HPCs or the land they occupy. However, upon using these models, one will further see that even in the worst case, interesting events happen in the best case. Meanwhile, the second model shows the goal of the entire section in its essence. The second inequality essentially says that the amount of power generated by renewable resources must be better or equal to the amount of non-renewable resources to ensure a 100% renewable source of energy for HPCs.

To better understand these models, the maximum possible s_{total} will be taken together with the minimum possible E_c . Considerations will be made by taking each type of renewable energy individually and together.

The following values will be assigned to each variable to better analyze the model:

Variables	Estimated Value
s_{total}	3716 square meters
E_c	0.5 MW
s_{solar}	1.6 square meters
s_{wind}	4045 square meters
C_{solar}	1 MW per 16,000 square meters
C_{wind}	4 MW per 200,000 square meters

Upon looking at these values, there is no need to integrate wind into any part of this model due to its size being too big to cover the place with minimal energy provided. Hence, the only part to be tried will be the solar panels.

By substituting the values,

$$n_{solar}(1.6 \text{ square meters}) \leq 3716 \text{ square meters} \Rightarrow n_{solar} \leq 2322.5$$

For the given HPC size, a maximum of 2322 solar panels can be implemented.

By substituting the value to the second inequality,

$$\left(\frac{1 \text{ MW}}{16,000 \text{ square meters}}\right)(2322 \text{ square meters}) \geq 0.5 \text{ MW} \Rightarrow 0.145235 \text{ MW} \geq 0.5 \text{ MW}$$

As demonstrated, the derived model resulted in a false statement even while considering the worst case of energy (the lowest possible) and the best case of the area (the largest possible at present). Hence, this model demonstrates the effects of transitioning into a 100% renewable energy source together with the challenges involved in implementing it. Even if all other factors are considered in this model, the lack of land area ultimately limits the capacities of renewable resources in the context of solving the negative environmental impact of HPCs.

Interestingly, the false statement provides insight into the maximum amount of renewable resources that can be utilized in all cases of HPCs. Here, by getting the ratio of the calculated value and the lowest value of power, the optimal percentage of renewable energy for a given HPC facility at this time is obtained.

Thus, the optimal percentage for renewable energy is:

$$\frac{0.145235}{0.5} = 0.29047 \approx 30\% \text{ renewable energy}$$

2.4.2 Increasing the Proportion of Renewable Energy

While a 100% proportion of renewable energy is not possible as of present, the proportion of renewable energy sources for HPC facilities can still be increased to the optimal setting to reduce the amount of carbon emissions brought about by the rise of HPCs around the world.

We have previously established that the optimal proportion of renewable energy to be utilized is 30%. Furthermore, in previous models, we have noted that the weighted average emission factor (M) can be calculated as:

$$M = \sum_{i=1}^{N_{src}} (P_i m_i).$$

Hence, by this model, we obtain that:

$$M = (0.7)(x) + 0.3(y)$$

where x is the sum of all the emission factors of non-renewable sources and y is the sum of all the emission factors of renewable resources.

For the context of this demonstration, we will let x be the emission factor of coal while y will be the emission factor of solar panels. M will first be calculated using the derived model then it will be compared to M without renewable resources. Using the derived model:

$$M = 0.7(0.93) + 0.3(0.04) = 0.663 \text{ kg of CO}_2/\text{kWh}$$

Meanwhile, with only non-renewable energy in mind, the yield is:

$$M = 1(0.93) = 0.93 \text{ kg of CO}_2/\text{kWh}$$

By getting the ratio of these two values we arrive at:

$$\frac{M_{derived}}{M_{original}} = \frac{0.663}{0.93} = 0.7129032258 \approx 70\%$$

Thus, this outcome matches the expected. By increasing the amount of renewable sources by 30%, the overall environmental impact increases by 30% indicating a direct relationship between the two factors. At the same time, the total weighted emissions decreases by 30% thus indicating an indirect relationship between the two factors.

2.5 Air Quality with HPCs and Fossil Fuel Power Plants

With the current energy sourcing of HPCs, fossil fuel power plants (FFPP) emitting carbon dioxide are required to ensure the continuous functionality of HPCs. However, aside from carbon emissions, these power plants also emit particulate matter (PM) and dust. These emissions ultimately affect the air quality of the surrounding areas and communities.

By these, it is logical to extend the current models by completely introducing the air quality factor as it is already connected to carbon emissions and environmental effects.

The environmental effects of both HPCs and FFPPs can be measured by:

$$I_{total} = I_{HPC} + I_{FFPP}$$

where

$$I_{HPC} = N_{HPC}E_C M + N_{HPC}E_C P_m + N_{HPC}E_C D_m \Rightarrow I_{HPC} = N_{HPC}E_C (M + P_m + D_m),$$

and

$$I_{FFPP} = N_{FFPP}E_F M + N_{FFPP}E_F P_m + N_{FFPP}E_F D_m \Rightarrow I_{FFPP} = N_{FFPP}E_F (M + P_m + D_m)$$

Recall that the weighted emission of carbon (M) is given by $M = \sum_{i=1}^{N_{src}} (P_i m_i)$.

Similarly,

$$P_m = \sum_{i=1}^{N_{src}} (P_i P_{src})$$

where P_{src} is the PM emission factor from that source and

$$D_m = \sum_{i=1}^{N_{src}} (P_i D_{src})$$

where D_{src} is the dust emission factor from the same source.

By combining everything, we arrive at:

$$\begin{aligned}
 I_{total} = & N_{HPC} E_C \left(\sum_{i=1}^{N_{src}} (P_i m_i) + \sum_{i=1}^{N_{src}} (P_i P_{src}) + \sum_{i=1}^{N_{src}} (P_i D_{src}) \right) \\
 & + N_{FFPP} E_F \left(\sum_{i=1}^{N_{src}} (P_i m_i) + \sum_{i=1}^{N_{src}} (P_i P_{src}) + \sum_{i=1}^{N_{src}} (P_i D_{src}) \right)
 \end{aligned}$$

This given model extends on the previously established models by primarily considering the PM emissions and dust emissions of the HPC and FFPP units together with the carbon emissions of these facilities. By creating this model, it directly connects to the amount of energy utilized by each HPC and power plant. Thus, the environmental impact of both these facilities together with the functionalities of each is obtained.

3 | Recommendations

3.1 Reducing Environmental Impact

There are a multitude of ways that HPC can become more environmentally friendly while also delivering the required computing power. The ideal step would be to transition to a 100% renewable energy grid to power HPC facilities, but this is unfeasible. Instead, HPC can adopt an energy model that runs on 30% renewable energy and 70% non-renewable energy which aligns with budgetary and resource considerations.

This means the optimal values for P_r and P_n , where these are proportions of renewable and non-renewable energy, are 0.3 and 0.7, respectively. Including this in our model,

$$M_r = \sum_{i=1}^{N_{src}} (P_{ri} m_{ri})$$

$$M_n = \sum_{i=1}^{N_{src}} (P_{ni} m_{ni})$$

where the subscripts indicate if the variable refers to renewable or non-renewable energy sources.

Finally,

$$M = 0.3M_r + 0.7M_n$$

Some technical methods to reduce energy consumption or utilize this energy more effectively can be adopted by HPC facilities for the same goal.

Another recommendation is to maximize the average utilization rate of HPC. If a power plant is outputting a certain amount of energy and the facility it powers is only consuming a fraction of it for computational tasks, then the remaining unused energy is funneled into less important components such as lighting the facility, among others. This unused energy could also go to waste but would still contribute to harming the environment.

Other than transitioning to renewable energy, a good option would be to invest in more energy-efficient components for HPC facilities. This allows for more operations to be done while consuming less energy. An example of this is switching to liquid cooling from air cooling, which reduces energy consumption by around 90%.

3.2 Letter to the United Nations Advisory Board

November 20, 2024

Mr. António Guterres

Secretary-General

United Nations

405 East, 42nd Street New York, NY, 10017, USA

Dear Mr. António Guterres:

Greetings!

We hope this letter finds you in good health and high spirits. We are a team of high school students who are writing to you — along with the United Nations Advisory Board — to address our concern regarding the lack of thorough discourse regarding the issue of High Powered Computing (HPC) facilities in the recent report entitled *Governing AI for Humanity*, particularly the HPCs' impact on our environment. This is especially shocking as, according to the 13th Sustainable Development Goals of the United Nations, urgent action to combat climate change and its impact is imperative to protect and transform our world into a healthy, just, and prosperous one. In addition, the fact that artificial intelligence (AI) heavily relies on these HPCs for further development, training, and utilization is another fact that remains relatively ignored in said report.

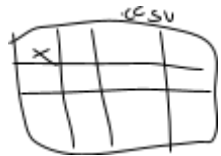
HPC systems power the most computationally demanding AI workloads, from training natural language processing models on huge amounts of data to simulations of large and complex systems (such as weather simulations). While the capabilities of these HPC facilities to perform these calculations are undeniably huge, so is their negative environmental impact. Together with the rapid development and deployment of AI technologies, HPC facilities have raised highly valid concerns about the sustainability of our digital advancements.

Henceforth, we have devised a mathematical model that is capable of predicting trends by analyzing, and thus quantifying, the environmental impact that HPC facilities have. Through research and mathematics, we have shown that by 2030, their environmental impact will be double that of 2024 — within six (6) years, the already alarming environmental impact of HPC facilities will double in magnitude. In addition, our model highlights the need for the transition to renewable and sustainable energy sources, energy-efficient components and algorithms, and the full utilization of existing HPC facilities to maximize already-available resources, and generate less waste. Therefore, these needs can only really be addressed through the advocacy for

transparent environmental impact reporting (done systematically with the help of our model), and global cooperation for all to take charge in this grave global issue.

Once again, we respectfully urge you to include these considerations in your upcoming developmental goals. Thank you for your commitment to governing AI for humanity, and we hope this plea contributes to your noble mission.

Sincerely,



Team .CCSV
Control No. 16302

4 | References

Statista. (2024, May 22). *Global number of internet users 2005-2023*.

<https://www.statista.com/statistics/273018/number-of-internet-users-worldwide/>

High Performance Computing Market Size, share & Trends Analysis Report by component (Servers, storage, networking devices), by deployment (On-premise, cloud), by end-use, by region, and segment Forecasts, 2023 - 2030. (n.d.).

<https://www.grandviewresearch.com/industry-analysis/high-performance-computing-market>

Development over Time | TOP500. (n.d.). <https://top500.org/statistics/overtime/>

HPC Cluster. (n.d.). Hewlett Packard Enterprise Development LP.

https://www.hpe.com/emea_europe/en/what-is/hpc-clusters.html

Nnene, O. A., Senshaw, D., Zuidgeest, M. H., Hamza, T., Grafakos, S., & Oberholzer, B. (2023). Baseline scenario modelling for low emissions development in Ethiopia's energy sector. *Energy Strategy Reviews*, 49, 101166. <https://doi.org/10.1016/j.esr.2023.101166>

High-Performance computing data center cooling system energy efficiency. (n.d.). Computational Science | NREL. <https://www.nrel.gov/computational-science/data-center-cooling-system.html>

Bulpit, K., & Bulpit, K. (2022, September 28). Why Liquid Cooling is the Most Eco-Friendly Solution for Data Centres - 101 Data Solutions. *101 Data Solutions - Data Management for Business*.

<https://101datasolutions.co.uk/why-liquid-cooling-is-the-most-eco-friendly-solution-for-data-centres/5698/>

How much energy do solar panels produce for your home | YES Energy Solutions. (n.d.).

<https://www.yesenergysolutions.co.uk/advice/how-much-energy-solar-panels-produce-home>

Turczyn, C. (2020, August 10). *Take a virtual tour of ORNL's Supercomputer Center – Oak Ridge Leadership Computing Facility*.

<https://www.olcf.ornl.gov/2020/08/10/take-a-virtual-tour-of-ornls-supercomputer-center/>

Wind Turbines: the Bigger, the Better. (n.d.). Energy.gov.

<https://www.energy.gov/eere/articles/wind-turbines-bigger-better>

How much energy do solar panels produce for your home | YES Energy Solutions. (n.d.).

<https://www.yesenergysolutions.co.uk/advice/how-much-energy-solar-panels-produce-home>

Wind energy. (n.d.). <https://www.irena.org/Energy-Transition/Technology/Wind-energy>